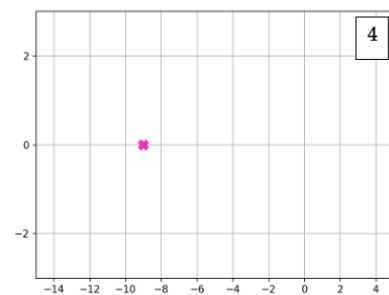
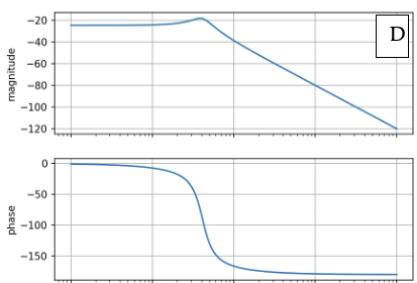
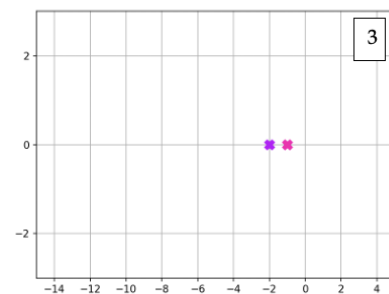
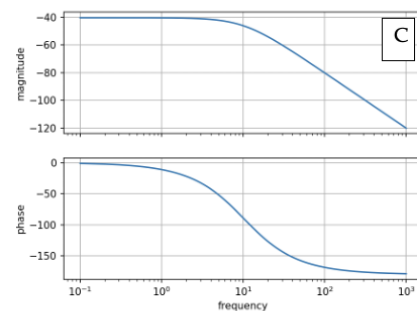
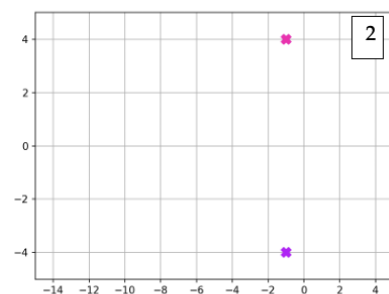
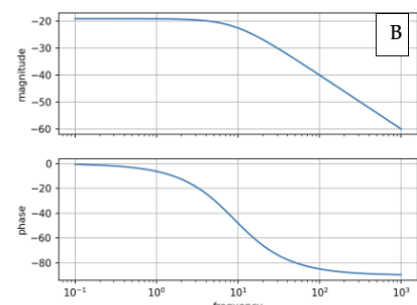
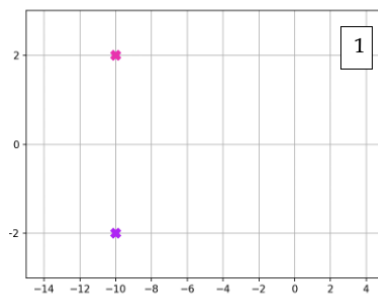
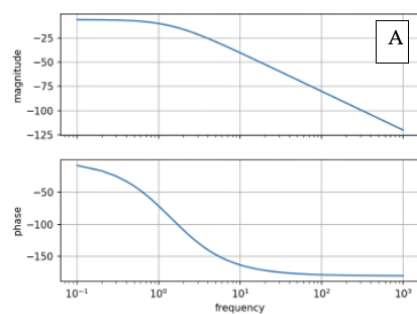


University of Cape Town
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EEE3094S - Control Systems Engineering

Problem Set 3.1 Bode Plots Solutions

Question 1

Match the following Bode plots to the poles and zeros of the system:



SOLUTION

- (a) A - 3 $\omega_0 = 1$ and $\omega_0 = 2$
- (b) B - 4 20 dB per decade roll off - 1 pole
- (c) C - 1 $\omega_0 = 10.19$
- (d) D - 2 Low damping leads to resonance bump at $\omega_0 \approx 4$

Question 2

Sketch rough Bode plots for the following systems:

(a) $\frac{1}{s + 0.1}$

(b) $\frac{s + 5}{s + 50}$

(c) $\frac{1}{s^2 + 16s + 68}$

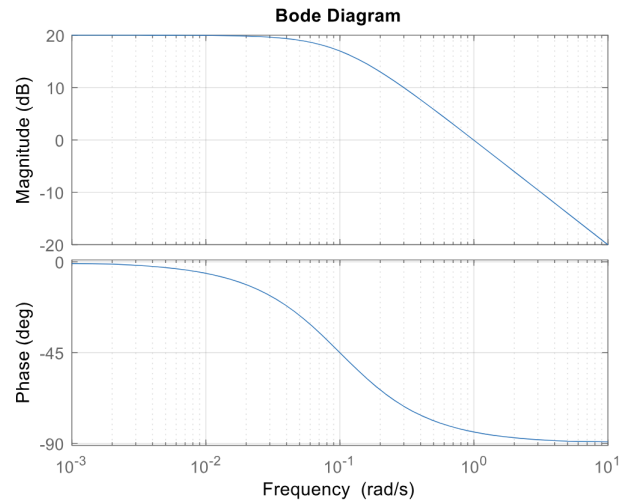
(d) $\frac{1}{(s + 10)(s + 12)(s + 20)}$

SOLUTION

(a) $\frac{1}{s + 0.1}$

$$\frac{1}{s + 0.1} = \frac{10}{10s + 1}$$

$$\omega_0 = 0.1, \quad A = 20$$



(b) $\frac{s + 5}{s + 50}$

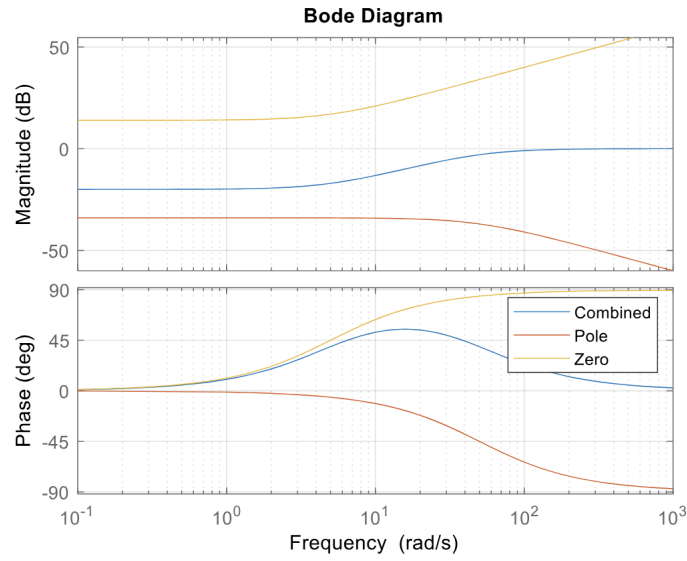
$$\frac{s + 5}{s + 50} = \frac{1}{s + 50}(s + 5) = \left(\frac{1}{50}\right) \left(\frac{1}{\frac{1}{50}s + 1}\right) \left(\frac{1}{5}s + 1\right)$$

Pole: $A = -34$ dB, $\omega_0 = 50$, Phase = -90°

Zero: $A = 14$ dB, $\omega_0 = 5$, Phase = $+90^\circ$

Before 5: $A = -20$ dB

After 50: $A = 0$ dB



(c) $\frac{1}{s^2 + 16s + 68}$

Rearranging to the standard second-order form:

$$\omega_n = 2\sqrt{17} = 8.24, \quad \zeta = \frac{16}{2\omega_n} = 0.97, \quad A = \frac{1}{68}$$

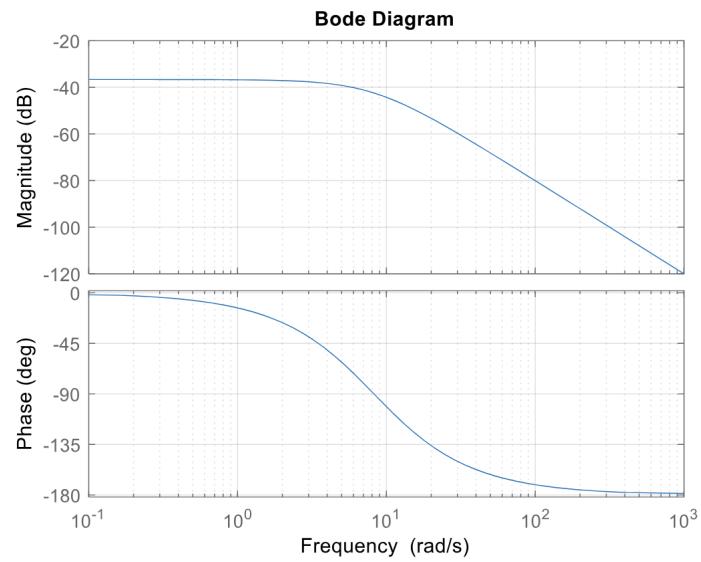
Poles at $-8 \pm 2j$

$$\omega_p = 2$$

Magnitude:

$$40 \log_{10} \left(\frac{1}{\sqrt{\sigma^2 + \omega_p^2}} \right) = -36 \text{ dB}$$

Important points: Initial magnitude, roll-off, and values at ω_n .

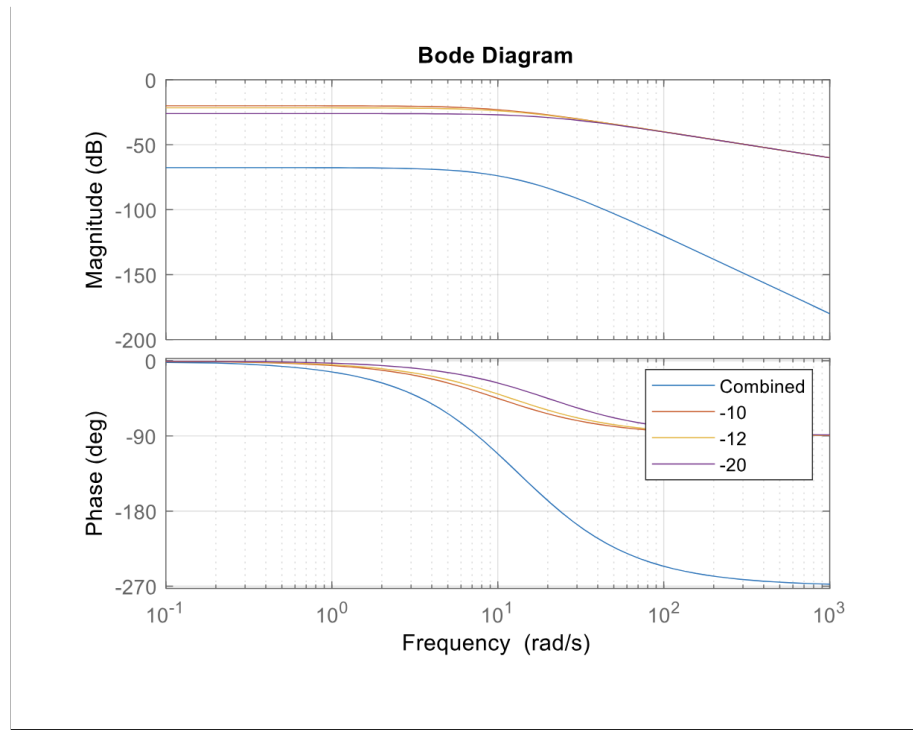


(d) $\frac{1}{(s + 10)(s + 12)(s + 20)}$

Poles:

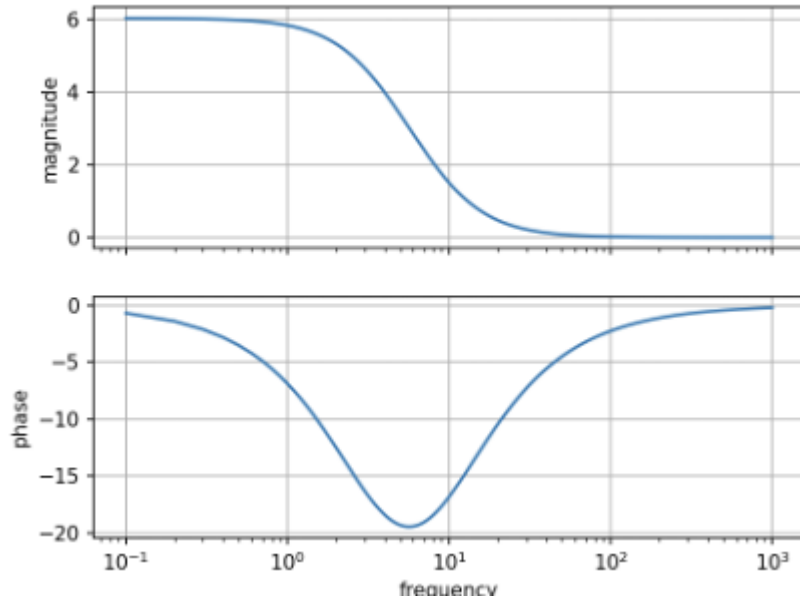
- At -10: $A = -20$ dB, $\omega_0 = 10$
- At -12: $A = -21.5$ dB, $\omega_0 = 12$
- At -20: $A = -26$ dB, $\omega_0 = 20$

Roll-off: -60 dB per decade. Find each critical point and subtract them



Question 3

The following Bode plot shows the frequency response of a first-order system.



- Does this system have a zero, and if so, is it to the right or left of the pole? Explain.
- What is the gain of the transfer function?
- Would you recommend using this system in an environment where it will be exposed to high-frequency noise? Explain.

SOLUTION :

- Does this system have a zero, and if so, is it to the right or left of the pole?**

The system has a zero because the phase returns to 0. The zero is faster than the pole (phase decreases then increases), therefore it is left of the pole.

- What is the gain of the transfer function?**

$$6 = 20 \log_{10} A \Rightarrow A = 2$$

- Would you recommend using this system in an environment with high-frequency noise?**

No. High-frequency signals are passed with no attenuation by this transfer function. They are not attenuated, and thus the system

would be sensitive to noise. The amplified low frequency may help with stability, but noise rejection is poor.